Qingqing Li

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EDUCATION

Master of Medicine Xuzhou Medical University, China **Sep 2020 - Jun 2023**

- GPA: 86.30/100
- Thesis: *Microglial Lysosomal Mechanisms for Major Depressive Disorder Alleviation*

Bachelor of Medicine North Sichuan Medical College, China **Sep 2015 - Jun 2020**

- GPA: 80.06/100 (range: 38/197)

RESEARCH EXPERIENCE

Research Assistant

Project 1: Cell-Type-Specific Mechanisms of Schizophrenia

Shanghai Mental Health Center, Shanghai, China **Jan. 2026 – Apr. 2026**

Supervisor: Dr. Jiaoqiong Guan

- Performed integrative single-cell bioinformatics analyses (*e.g.*, hdWGCNA, scFEA, scTranslator) and validation experiments using the schizophrenia mouse model.
- Identified the role of the target gene in conferring disease susceptibility within distinct neuronal populations, offering insights into cell-type-specific mechanisms underlying schizophrenia.

Research Assistant

Project 2: Cross-Disorder Genetic Architecture and Single-Cell Functional Mapping

Shanghai Pudong New Area Mental Health Center, Tongji University School of Medicine, Shanghai, China **Jun. 2025 – Mar. 2026**

Supervisor: Dr. Weijie Xie

- Conducted integrative genetic analyses by combining GWAS data with Mendelian Randomization, Transcriptome-Wide Association Study (TWAS), and Bayesian colocalization, along with single-cell analysis (e.g., scINSIGHT).
- Constructed a cross-disorder, cross–brain region genetic causal map showing genes across disorders, with follow-up single-cell analyses revealing both shared and distinct functional features.

Research Assistant

Project 3: Microglia-GABAergic Neuron Crosstalk in Major Depressive Disorder

Shanghai Pudong New Area Mental Health Center, Tongji University School of Medicine, Shanghai, China **Aug. 2024 – May. 2025**

Supervisor: Dr. Weijie Xie

- Utilized single-cell RNA sequencing data analysis (*e.g.*, CellChat, pySCENIC, Monocle) integrated with Mendelian Randomization and Machine Learning models (*e.g.*, random forest, SVM).
- Identified impaired microglia subtypes characterized by metabolic dysfunction and chronic neuroinflammation, which drive aberrant microglia-GABAergic neuron crosstalk in Major Depressive Disorder.

Research Assistant

Project 4: Comorbidity Mechanisms of Autoimmune and Psychiatric Disorders

Shanghai Mental Health Center, Shanghai, China **Jul. 2023 - Present**
Supervisor: Dr. Jiaoqiong Guan

- Performing integrative bioinformatics analyses (*e.g.*, WGCNA, Machine Learning, single-cell analyses) and validation experiments using the Chronic Unpredictable Mild Stress model.
- Aiming to identify an immune target that may underlie the shared immune pathogenesis which drives the comorbidity progression of Systemic Lupus Erythematosus and Major Depressive Disorder.

Master's Student
Project 5: Lysosomal Mechanisms of Microglia in Depressive-like Behaviors
Xuzhou Medical University, Xuzhou, China **Sep. 2020 – Jun. 2023**
Supervisor: Prof. Dr. Wuyang Wang

- Applied *in vivo* (LPS / Chronic Restraint Stress mouse model, stereotaxic surgery) and *in vitro* (primary microglia isolation and culture) techniques.
- Characterized a novel microglial lysosomal molecular mechanism implicated in the pathophysiology of depressive-like models and proposed a potential targeted intervention.

PUBLICATIONS

- **Qingqing Li,** *et al*. Mendelian Randomization Analysis of Multiregional Psychiatric Disorders. **(Submitted)**
- **Qingqing Li**, *et al*. Single-Cell Dissection of Microglial Heterogeneity Reveals Subtype-Driven Immune-Metabolic Crosstalk in Major Depressive Disorder. *Current Neuropharmacology*. **(Accepted, JCR Q1)**
- **Qingqing Li**, *et al*. Microglia sing the prelude of neuroinflammation-associated depression. *Molecular Neurobiology*. 2025 Apr;62(4):5311-5332. **(JCR Q1)**
- Yanhong Xing, Meng-Meng Wang, Feifei Zhang, Tianli Xin, Xinyan Wang, Rong Chen, Zhongheng Sui, Yawei Dong, Dongxue Xu, Xingyu Qian, Qixia Lu, **Qingqing Li,** *et al*. Lysosomes finely control macrophage inflammatory function via regulating the release of lysosomal Fe^{2+} through TRPML1 channel. *Nature Communications*. 2025 Jan 24;16(1):985. **(JCR Q1)**
- Jiansong Qi, **Qingqing Li,** *et al*. MCOLN1/TRPML1 in the lysosome: a promising target for autophagy modulation in diverse diseases. *Autophagy*. 2024 Aug;20(8):1712-1722. **(JCR Q1)**

SKILLS

- **Wet-Lab Techniques**: qPCR, Western Blot, Immunofluorescence/Immunohistochemistry, ELISA, Golgi Staining, Gel Electrophoresis, Cell Culture, Animal Models.
- **Programming languages:** R, Python.
- **Bioinformatics:** Single-cell Analysis, Spatial Transcriptomics, DNA Methylation Analysis.
- **Statistical Genetics**: Mendelian Randomization, TWAS/PWAS, Colocalization.
- **Operating System**: Linux (command line, scripting).
- **Software:** EndNote, GraphPad Prism, ImageJ, Imaris, Adobe Illustrator & Photoshop.
- **Languages:** Mandarin Chinese (Native), English (**IELTS 6.5**).

ACADEMIC SERVICE

- Peer Reviewer for ***Molecular Neurobiology***, ***Cellular and Molecular Neurobiology,*** and ***Scientific Reports*****.**

2024-Present

HONORS

- Postgraduate Scholarship, Xuzhou Medical University. **2020-2023**

REFERENCES

- Prof. Dr. **Wuyang Wang** (Master's Thesis Supervisor) | wuyangwang80@gmail.com
- Prof. Dr. **Jiaoqiong Guan** (Research Supervisor) | guanjiaoqiong@kmmu.edu.cn
- Prof. Dr. **Weijie Xie** (Research Supervisor) | xwjginseng@126.com